

MOHAMMAD ASIF ZAFAR

Senior Android Engineer | 8+ yrs Kotlin, Jetpack Compose, MVVM | Coroutines, RxJava, Modular Apps | Secure, Scalable, Customer-Focused Mobile Solutions

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Professional Summary

Senior Android Engineer with 9.5 yrs of experience building and scaling Kotlin-based Android apps using Jetpack Compose, MVVM, and modular architectures. Skilled in Coroutines, Flow, and RxJava for reactive programming, with expertise in secure app development (SSL pinning, PCI DSS, fraud detection) and CI/CD pipelines. Proven track record delivering customer-facing apps to the Play Store in cross-functional Agile squads, with strong focus on product outcomes, clean code, and mentoring Android teams.

Improved test coverage from 45% → 85%, reducing QA cycles by 40% with automated testing frameworks (JUnit, Espresso, Mockito).

Contributed to apps serving 10M+ users across 4 countries, ensuring scalability, security, and WCAG accessibility compliance.

Core Skills

- **Languages & Frameworks:** Kotlin, Jetpack Compose, Kotlin Multiplatform (KMP), Compose Multiplatform (CMP), Coroutines, Async/Await Networking, MVVM, Clean Architecture, Hilt/Dagger, Retrofit, REST APIs, GraphQL APIs, Java, C++ (basic) with JNI integration, Google Play Services, Material Design 3, Android 10+ compatibility, modular & multi-module architecture, EMV payment workflows, WCAG accessibility compliance, customer-facing enterprise apps, real-time data integration, IoT connectivity.
- **DevOps & Tooling:** Jenkins, CI/CD pipelines (GitHub Actions, GitLab CI, Bamboo, Bitbucket), Play Console, SonarQube, Android Studio, ADB, Maven, Fastlane, Firebase, Jira, artifact management.
- **Testing & Automation:** JUnit, Mockito, Espresso, Barista, Snapshot testing, Turbine, Perfecto (scriptless testing), UI/E2E testing, TDD, Mutation testing, automated testing infrastructure for enterprise-grade apps.
- **APIs & Cloud Platforms:** Retrofit, REST, GraphQL, ThreatMetrix, Google Maps SDK, Spring Boot, AWS, GCP, Azure, microservices architecture, SaaS & private cloud integration, MDM compatibility.
- **Specialist Expertise:** AI-driven feature integration (GenSearch-ready), mobile performance & scalability optimization, secure architecture (SSL pinning, session timeout, encrypted local storage), analytics instrumentation, advanced debugging, pair programming, Agile (Scrum/Kanban), team mentorship & technical guidance (led & coached 6+ developers), API documentation/specification, PCI DSS compliance, secure SDK development, third-party library selection & integration (well-maintained, latest SDK compatibility).
- **Also Familiar With:** Realm database, Postman, Charles Proxy/Proxyman, modular feature/library architecture, release management (weekly cycles), UI/UX collaboration, component library integration.
- **Technical Exploration:** Exploring Rust for cross-platform business logic and memory-safe concurrency (ownership model, FFI via Kotlin/Native, Cargo ecosystem).

Professional Experience

UOB Bank (via Nityo → iTAC) | Kotlin, Java, Jetpack Compose, Retrofit, Hilt/Koin, Espresso, Jenkins, Perfecto, SonarQube

Senior Android Developer | Malaysia – Jan 2023 – Present

- **Modular Architecture & Compose Integration:** Built scalable, modularized Android 10+ architecture using **Kotlin, Jetpack Compose, MVVM, and Coordinators**, aligning with **Google Android architecture guidelines** and enabling feature-driven delivery. Delivered customer-facing digital banking features with an emphasis on performance optimization, **secure real-time data handling**, and resource efficiency principles applicable to **energy-data workflows** and **grid-aware mobile apps**.
- **Client-Facing Delivery & Collaboration:** Collaborated in **cross-functional Agile squads** with Product Managers, Designers, and Backend Engineers to define **API contracts, refine features, and align delivery** with product goals and customer outcomes.
- **Secure App Flows:** Integrated ThreatMetrix to analyze 150+ real-time fraud indicators; enforced SSL pinning, encrypted storage, and secure onboarding flows — leveraging the same real-time event processing and secure API integration methods used in smart energy monitoring, renewable energy platforms, and **EMV-compliant digital payment workflows**.
- **Testing Strategy & Automation:** Enhanced **test coverage from 45% → 85%** with **JUnit, Mockito, Espresso, and Haplo-style snapshot tests**, improving release stability and reducing manual QA regression effort by 40% and supporting stable weekly CI deployments.
- **Code Quality & Monitoring:** Maintained strong code health using SonarQube, GitHub Actions, and static analysis; mitigated crashes and performance issues through proactive monitoring with Firebase Crashlytics and Android Vitals.
- **Mentorship & Agile Practices:** Conducted code reviews, onboarded junior engineers, and introduced reusable, accessible (WCAG-compliant) UI components—enhancing delivery velocity and maintainability across squads.
- **CI/CD & Release Ownership:** Managed release pipelines using Jenkins, GitHub Actions, and Bamboo; defined Git workflows and quality gates, ensuring production-ready builds and smooth app store submissions across regions.

Tata Consultancy Services (Star Alliance) | Java, Kotlin, Android Studios, Amazon AWS Pipeline, S3 bucket

Senior Android developer | India– Apr 2021 – Dec 2022

- **Modular Development of Frequent Flyer App:** Developed and maintained **customer-facing global Android apps** with Kotlin, MVVM, and Jetpack libraries, ensuring **Android 10+ compatibility** and delivering scalable, high-quality features through close collaboration with **designers and backend engineers**.
- **Jetpack Navigation & Deep Linking Integration:** Implemented deep link navigation using Jetpack Navigation Component and supported partner airline linking, enhancing user onboarding across 26+ member airlines.
- **Offline-First Architecture with Room DB:** Enabled offline access to loyalty data using Room and WorkManager, increasing daily engagement for frequent travelers in low-connectivity environments.
- **Localization & Globalization:** Delivered multilingual support (10+ languages) with dynamic resource handling, enabling Star Alliance app expansion across APAC, EU, and LATAM.
- **Accessibility & Compliance Enhancements:** Integrated accessibility best practices (TalkBack, dynamic font sizing), aligning with WCAG 2.1 standards and expanding inclusivity for users with disabilities.
- **Cross-Team Collaboration with Remote Squads:** Coordinated with cross-functional teams in Germany and India to align feature delivery, leading to a 20% improvement in release predictability.
- **Developed reusable Android SDK modules** for airline integrations, driving consistency, scalability, and reducing time-to-market for **26+ partner airlines**.

Adaptive Control Security Global | Kotlin, Java, Android Studio, PHP

Executive Android Developer | India | Feb 2018 – Jan 2021

- **Custom Security App Development:** Delivered **MVVM/Clean Architecture Android apps in Kotlin**, integrating **real-time alerts, offline-first data handling, and QR scanning**, improving operational response time by 50%.
- **Real-Time Alerts & QR Code Integration:** Integrated QR code scanning and real-time notifications with Firebase Cloud Messaging (FCM), reducing manual entry errors and streamlining checkpoint workflows.
- **Local Storage with Realm Database:** Leveraged Realm for offline data storage, enabling secure access control logs in remote facilities without network access.
- **Security Camera Integration & Remote Access:** Enabled live streaming from IP cameras using ExoPlayer and WebRTC, improving real-time monitoring capabilities for clients across multiple sites.
- **CI/CD with Bitrise:** Configured Bitrise for automated builds and testing, reducing manual QA effort and enabling faster turnaround on client feature requests.
- **Stakeholder Collaboration for Product Delivery:** Engaged with internal business stakeholders and operational leads to gather feature requirements, drive roadmap alignment, and support digital transformation efforts across enterprise banking modules.

Inteq Infocom Solution Pvt. Ltd. | Java, Android Studio

Android Developer | India | Jun 2016 – Oct 2017

- **e-Governance App Development :** Developed a digital inspection Android app using Java and SQLite, digitizing daily site reports and increasing reporting efficiency by 60%.
- **Offline Sync with SQLite and REST APIs:** Implemented background sync logic and data caching, enabling field officers to work offline and sync data automatically when reconnected, reducing data loss incidents.
- **User Authentication and Secure Access:** Implemented **secure login flows with encrypted local storage, OTP validation, and session timeout**, strengthening data protection and user trust in government e-governance apps..
- **Modular UI Design with Custom Views:** Designed reusable UI components for forms and checklists, improving UI consistency and reducing development time for new features.
- **Collaboration with Stakeholders:** Worked directly with state and central railway departments to gather requirements and deploy updates, helping to drive digital transformation at scale.

Projects

- **UOB TMRW & Mighty (MY, SG, TH, ID):** Multi-country banking features with Kotlin, Retrofit, Jetpack Compose, Threat Metrix, and CI/CD via Jenkins & Bitrise..
[TMRW Indonesia](#) | [TMRW Thailand](#) | [UOB Mighty MY](#) | [UOB Mighty SG](#)
- **Star Alliance Baggage Hub (TCS):** Baggage tracking and scanning app for Lufthansa & Singapore Airlines using Scandit SDK and AWS backend. [Star Alliance Baggage Hub](#)
- **Fujifilm Healthcare IT:** Designed Android UIs for medical workflow tools, ensuring usability and high-performance rendering for clinical settings.
- **AT&T Cricket SDK App:** Integrated Cricket SDK and APIs to develop optimized Android apps for carrier-specific devices. [Cricket App Launcher](#)
- **Sprint Voicemail (T-Mobile):** Enhanced voicemail functionality for pre-OEM and BYOD Android devices, improved security and login persistence
- **MedBuddy (Open Source)** – Modern health tracking app with **offline-first architecture**, built using **Kotlin Compose Multiplatform** and **Jetpack Compose**. [Github/MedBuddy](#)
- **Ramadan Kareem (Open Source)** – Modern Islam App with **first architecture**, built using **Kotlin Multiplatform** and **Jetpack Compose**. [Github/Ramadan kareem](#)

Education

B. Tech, Computer Engineering – Punjab Technical University, Punjab (2012–2016)

Senior Secondary – St. Edmund's School, Jaipur (2009–2011)

Secondary – St. Andrew's School (2000–2009)